



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

P2 Electricity: Test Review Checklist

Test: Wednesday 7 October, Period 4

Everything on the test is on this page. Tick a box when you can do it without looking at your notes.

1. Ohm's Law and Circuits

- ☐ Draw and read circuit diagrams with all the standard symbols. Ammeter in **series**, voltmeter in **parallel**.
- ☐ Current is the rate of flow of charge, $Q = I t$. It is the same at every point in a single loop.
- ☐ **Ohm's law**: use and rearrange $V = I R$. For the same potential difference (voltage), more resistance means less current.
- ☐ **Series circuits**
 - ☐ The same current through every component.
 - ☐ The supply potential difference (voltage) is shared between the components.
 - ☐ Resistances add: $R_{\text{total}} = R_1 + R_2$. Calculate current, p.d. and resistance.
- ☐ **Parallel circuits**
 - ☐ The same p.d. across each branch.
 - ☐ The branch currents add up to the total current.
 - ☐ Total resistance is less than the smallest resistor. Explain why series raises total resistance and parallel lowers it.
- ☐ **Power**: use $P = V I$ and $P = I^2 R$.
- ☐ **Energy**: $E = P t$ and $E = Q V$ — learn these two from the review packet, **Q10 and Q12**. An appliance transfers more energy the higher its power and the longer it is on.

2. Required Lab Practicals

- ☐ **RP15 — Resistance of a wire**
 - ☐ The circuit: ammeter in series, voltmeter across the wire, crocodile clips on the wire along a metre rule.
 - ☐ Change the length, record V and I , calculate $R = V \div I$. Keep the current small so the wire stays at a constant temperature.
 - ☐ Plot R against length: a straight line through the origin, so resistance is directly proportional to length. Also: resistors in series and in parallel.
- ☐ **RP16 — I-V characteristics**
 - ☐ The circuit: variable resistor to change the current, ammeter in series, voltmeter across the component.
 - ☐ Take readings, then reverse the supply to get the negative values.
 - ☐ Plot I against V for a resistor, a filament lamp and a diode.

3. Ohmic and Non-Ohmic

- ☐ **Ohmic** (a resistor at constant temperature): current is directly proportional to p.d., a straight line through the origin; resistance stays constant.
- ☐ **Non-ohmic**: filament lamp (resistance rises as it heats) and diode (current one way only, very high resistance in reverse). Sketch and explain all three I-V graphs.
- ☐ Find resistance at a point on a graph: $R = V \div I$.

4. Electrical Grid

- ☐ **ac and dc**: direct p.d. stays in one direction; alternating p.d. keeps reversing. UK mains is ac, about **230 V** at **50 Hz**.
- ☐ **Mains**: three-core cable — live (brown, about 230 V), neutral (blue, about 0 V), earth (green and yellow, 0 V, carries current only in a fault) — and the job of each. Explain why the live wire is dangerous even with the switch off.
- ☐ **National Grid**: cables and transformers; step-up for the cables, step-down for homes. Explain why it is efficient: higher p.d., smaller current, less heating ($P = I^2 R$).
- ☐ **Higher tier**: use $V_p I_p = V_s I_s$ (given on the equation sheet).